

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
Weightage	25%	25%	25%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **J.RAMESH(192311230)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ramesh

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID Symptoms Observed

P1 Fever, Cough
P2 Fever, Headache
P3 Cough, Headache
P4 Fever, Cough, Headache
P5 Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "**Fever**". Identify the valid association rules and justify why they satisfy the given constraints.

Q2. Dataset: Online Learning Platform

Student ID Courses Enrolled

S1 Python, SQL
S2 Python, Data Science
S3 SQL, Machine Learning
S4 Python, SQL, Data Science
S5 Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

Q3. Dataset: Library Borrowing Records

Transaction ID Books Borrowed

T1 Data Mining, Python Programming

Transaction ID	Books Borrowed
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "**Computer Science**" level. Explain how the course hierarchy affects the discovery of association rules.

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Answers

Q1. Association Rule Mining – Hospital Patient Records

Given

- Total transactions = 5
- Minimum Support = 50%
- Minimum Confidence = 70%
- Constraint: Every rule must include Fever

Step 1: Find Support

Itemset	Count	Support	Frequent?
Fever	4	80%	Yes
Fever, Cough	2	40%	No
Fever, Headache	2	40%	No
Fever, Fatigue	1	20%	No
Fever, Cough, Headache	1	20%	No

Since the minimum support is 50%, only Fever (80%) is a frequent itemset.

Step 2: Check Association Rules

Rule 1: Fever $\rightarrow$ Cough

$$\text{Support} = \frac{2}{5} \times 100 = 40\%$$

$$\text{Confidence} = \frac{\text{Support}(\text{FeverCough})}{\text{Support}(\text{Fever})} = \frac{2}{4} \times 100 = 50\%$$

Invalid because support is $40\% < 50\%$ and confidence is $50\% < 70\%$.

Rule 2: Fever $\rightarrow$ Headache

$$\text{Support} = \frac{2}{5} \times 100 = 40\%$$

$$\text{Confidence} = \frac{2}{4} \times 100 = 50\%$$

Invalid because support $< 50\%$ and confidence $< 70\%$.

Rule 3: Fever $\rightarrow$ Fatigue

$$\text{Support} = \frac{1}{5} \times 100 = 20\%$$

$$\text{Confidence} = \frac{1}{4} \times 100 = 25\%$$

Invalid because support < 50% and confidence < 70%.

Final Answer

No valid association rules are obtained.

Although Fever occurs in 4 out of 5 patients, giving it 80% support, none of the combinations containing Fever with another symptom reaches the required 50% support. Therefore, no rule can satisfy both the 50% minimum support and 70% minimum confidence requirements.

Conclusion

Valid Association Rules = None

Frequent itemset = {Fever} with 80% support

Reason: All Fever-based combinations have support below 50%.

Q2. Frequent Itemset Mining – Online Learning Platform

Given

- Total students = 5
- Minimum Support = 40%
- Minimum support count:

$$5 \times 40\% = 2$$

Therefore, an itemset must occur in at least 2 students to be frequent.

- Constraint: Each itemset must contain at least 2 courses.

Step 1: 2-Course Itemsets

Itemset	Students	Count	Support	Result
Python, SQL	S1, S4	2	40%	☒ Frequent

Itemset	Students	Count	Support	Result
Python, Data Science	S2, S4	2	40%	Frequent
Python, Machine Learning	S5	1	20%	Not Frequent
SQL, Data Science	S4	1	20%	Not Frequent
SQL, Machine Learning	S3	1	20%	Not Frequent
Data Science, Machine Learning	None	0	0%	Not Frequent

Step 2: 3-Course Itemsets

The possible 3-course combinations are:

Itemset	Students	Count	Support	Result
Python, SQL, Data Science	S4	1	20%	
Python, SQL, Machine Learning	None	0	0%	
Python, Data Science, Machine Learning	None	0	0%	
SQL, Data Science, Machine Learning	None	0	0%	

No 3-course itemset satisfies the 40% minimum support.

Final Frequent Itemsets

Therefore, the frequent itemsets satisfying both constraints are:

1. **{Python, SQL}** → Support = **40%**
2. **{Python, Data Science}** → Support = **40%**

How the Constraint Helps in Pruning

The constraint “each itemset must contain at least two courses” eliminates all single-course itemsets from consideration.

For example:

- {Python} → not considered because it contains only one course.
- {SQL} → not considered.
- {Data Science} → not considered.
- {Machine Learning} → not considered.

Thus, the mining process can directly focus on 2-course and larger combinations, reducing the number of candidate itemsets and making the mining process more efficient.

Conclusion

Frequent itemsets:

{Python, SQL} – 40% support

{Python, Data Science} – 40% support

The minimum-size constraint helps prune unnecessary single-course candidates and reduces the search space during frequent itemset mining.

Q3. Library Borrowing Records – Apriori Algorithm

Given

- Total transactions = 5
- Maximum itemset size = 3
- Since no minimum support is specified, we consider itemsets occurring in at least 2 transactions (40%) as frequent.

Step 1: 1-Itemsets

Itemset	Count	Support
Data Mining	4	80%
Python Programming	3	60%
Database Systems	3	60%
Artificial Intelligence	1	20%

Frequent 1-itemsets:

{Data Mining}, {Python Programming}, {Database Systems}

Step 2: 2-Itemsets

Itemset	Count	Support	Result
Data Mining, Python Programming	2	40%	Frequent
Data Mining, Database Systems	2	40%	Frequent
Python Programming, Database Systems	2	40%	Frequent

Therefore:

$L_2 = \{\text{DM, Python}\}, \{\text{DM, Database}\}, \{\text{Python, Database}\}$

Step 3: 3-Itemsets

The three frequent items form:

{Data Mining, Python Programming, Database Systems}

It occurs only in T4.

$$Support = \frac{1}{5} \times 100 = 20\%$$

Therefore, it is not frequent under the assumed 40% support threshold.

Final Frequent Itemsets

- {Data Mining} → 80%
- {Python Programming} → 60%
- {Database Systems} → 60%
- {Data Mining, Python Programming} → 40%
- {Data Mining, Database Systems} → 40%
- {Python Programming, Database Systems} → 40%

Effect of Maximum Size = 3

The constraint prevents Apriori from generating itemsets containing 4 or more books. Therefore, candidate generation stops after the 3-itemset level. This significantly reduces the search space, memory usage, and number of support calculations.

Conclusion: The maximum itemset-size constraint improves Apriori efficiency by limiting candidate generation to itemsets of size 1, 2, and 3.

Q4. University Course Hierarchy – Multilevel Association Rules

Given

The course hierarchy contains courses such as:

- Engineering
- Computer Science
 - Machine Learning
- Information Technology

The constraint says that rules should be generated only at **the Computer Science level**.

Students taking Computer Science:

- S1 → Computer Science, Machine Learning
- S3 → Computer Science
- S4 → Computer Science, Machine Learning

Frequent Patterns at Computer Science Level

Itemset	Count	Support
Computer Science	3	60%
Computer Science, Machine Learning	2	40%

Thus, the meaningful association is:

Computer Science → Machine Learning

$$Support = \frac{2}{5} \times 100 = 40\%$$

If confidence is calculated:

$$Confidence = \frac{Support(C, SML)}{Support(CS)} = \frac{2}{3} \times 100 = 66.67\%$$

So the rule indicates that approximately 66.67% of students who take Computer Science also take Machine Learning.

Effect of Course Hierarchy

Multilevel association mining allows relationships to be studied at different levels of abstraction. The constraint restricting the analysis to Computer Science eliminates unrelated course-level rules such as Engineering → Information Technology.

This reduces the search space and focuses the analysis on relationships relevant to Computer Science students.

Conclusion: The hierarchy helps identify meaningful relationships between a parent course area and its related specialization, such as Computer Science → Machine Learning.

Q5. Insurance Customer Records – Multidimensional Association Rules

Given Constraint

Age Group = Young

Young customers are:

Customer Income Policy

C1	High	Health Insurance
C3	Low	Travel Insurance
C5	Medium	Vehicle Insurance

Therefore, the analysis is restricted to 3 Young customers.

Patterns Identified

Age Group	Income Level	Policy	Customers
Young	High	Health Insurance	C1
Young	Low	Travel Insurance	C3
Young	Medium	Vehicle Insurance	C5

The observed patterns are:

1. Young + High Income → Health Insurance
2. Young + Medium Income → Vehicle Insurance
3. Young + Low Income → Travel Insurance

Each pattern occurs once among the three Young customers, giving:

$$\text{Support} = \frac{1}{5} \times 100 = 20\%$$

Within the Young-only subset, each pattern has:

$$Support = \frac{1}{3} \times 100 = 33.33\%$$

Marketing Applications

- Young, high-income customers can be targeted with Health Insurance plans and premium health coverage.
- Young, medium-income customers can be targeted with affordable Vehicle Insurance packages.
- Young, low-income customers can be offered low-cost and flexible Travel Insurance plans.

Conclusion

The constraint Age Group = Young reduces the dataset to the relevant customer segment. This makes it easier to identify income-based policy preferences and develop targeted marketing campaigns instead of promoting the same insurance product to all customers.